

1. Comparison between Analog and Digital System or Advantages, disadvantages of digital system (page-2, 4)	Lecture 1
2. Comparison between 1's and 2's compliment *** (page - 25,26) [the blue marked 3 points]	Lecture 1
Just remember Non-Weighted code - Doesn't follow the positional weight method like binary, 8421 or BCD - Example: Gray code, Excess-3 code etc Weighted code - follows the positional weight method - Example: binary, 8421 or BCD	Lecture 2
3. Conversion of Decimal to BCD Code (page - 4,5) BCD Addition *** (6 - 8)	Lecture 2
4. Limitations of BCD code *** - Needs extra correcting codes - Then Extra circuit is required for correcting - This extra circuit makes it slower	Lecture 2
4. Gray code properties *** (page - 12) Gray code conversion (Gray to Binary, Binary to Gray) *** (page - 13,14)	Lecture 2
5. Suppose 'X' is a decimal number composed of last two digits of your matric no and 'Y' is a decimal number composed of last two digits of your matric no. For example: If your metric no is C233093 then $X = (93)_{10}$ $Y = (23)_{10}$. i) Now convert 'X' and 'Y' into equivalent BCD codes and ii) convert each of the BCD codes into a corresponding Gray code. iii) add the converted BCD numbers using BCD addition ***	Lecture - 2

6. Reflective/Self-complementing code. 2421/3321/7421 is self-complementing/reflective code or not *** (Page 11)	Lecture - 2
7. Excess-3 code and prove that excess-3 is a self-complementing/reflective code*** (Page 9)	Lecture - 2
8. parity code *** (page 15) parity code disadvantages: • Parity Code Only Works for Single-Bit Errors. It will not detect multiple errors • It can only detect error and can't correct it	Lecture - 2
9. hamming code generation *** (page 18 - 19) or hamming code is given detect and correct errors (page - 20) ***	Lecture - 2
11. a) Convert Boolean expression in standard form **** $F=y'+xz'+xyz$. Minimize the following 5 variables Standard form means simplification. Solve this type of simplification problems from Boolean Algebra pdf (page 11) ****	Boolean Algebra pdf
12. State and Prove the De Morgan's Law of boolean Algebra with truth table and circuit diagram.*** (page 16 - 17)	Lecture - 3
13. 4-variable K-map with don't care*** 5- variable K-map*** (Solve previous year. You can get it from my playlist)	Lecture 4+5
14. $F(A,B,C,D) = \sum(1, 3, 5, 7, 11, 15)$ *** i. Simplify and obtain the boolean expression in SOP and ii. Implement the boolean expression using NAND gates only.	

14. Standard to Canonical form of SOP (page - 14) Standard to Canonical form of POS [Only the problem on the right side] (page - 15)	Lecture - 4
15. Write down the steps to design a Combinational circuit.*** (page 5)	Lecture - 6
16. Combinational circuit problems. [Problem 1, 3, 5, 6, 7] + Previous questions. [Can't be more specific at this stage. If possible I will inform later]	Lecture - 6
17. Half Adder***/ Design a circuit to add two single-bit binary values, A, B to produce a sum S and a carry out C. (page 6,7) (Only truth table, expression and the circuit with XOR gates only)	Lecture - 6
18. Design a Full adder using Half Adders*** (page 8 and 11)	Lecture - 6
19. Design a Binary Parallel Adder/Ripple carry adder (page 12, 13).	
20. Basic gate implementation using Nand/NOR Gates (page 31)	Lecture - 6
21. Nand/NOR Gate Implementation*** (page 37) Implement $A+B'C$ using Nand/NOR Gates only	Lecture - 6
22. Page (38 - 41) Consensus/Redundancy Theorem*** Stuck 1 and Stuck 0 [Circuit is not required] What is Self dual? [First 3 lines] *** What is the total number of self-duals of a function which has n variables? how many two input NAND gate required to implement 3 Input NAND gate.***	Lecture - 6